

Activity: Simulate Bayer Demosaicing

Introduction

Digital cameras capture images using a Bayer filter, where each pixel records only one color (Red, Green, or Blue). The missing colors must be reconstructed using a process called demosaicing, which estimates the values of neighboring pixels to create a full-color image.

This activity will guide you through simulating a Bayer filter array (RGGB pattern) and applying demosaicing techniques using OpenCV to reconstruct an image. You will explore how raw image data is processed to achieve a final, visually accurate color representation.

Learning Outcomes

By the end of this activity, you should be able to:

- Implement a basic demosaicing function using Python/OpenCV and visualize the reconstructed color image.

Working with Dev Container

To complete this activity in a reliable and fully configured environment, please refer to the instructions in the file: `README-devcontainer.md`. This guide walks you through opening the assignment in **Visual Studio Code** using a **Dev Container**, which automatically installs all necessary Python libraries and tools. Following that setup ensures that the notebook runs smoothly without manual configuration or missing dependencies. Make sure to open the **main activity folder** in VS Code and follow the steps outlined in the Dev Container guide before starting the notebook.

Instructions

Start by opening the Jupyter notebook in `Code` folder provided with this activity. This notebook is designed to help you **understand how a digital camera reconstructs a full-color image** from a sensor that only captures one color per pixel.

There will be tasks for you to complete as practice. These will be indicated by comments in the code block. In some cases, we will provide some starter code in a comment block. Other times we just provide a prompt for you to complete.

Working With Jupyter

With this type of environment, code can be executed out of order. That can have unintended consequences. Errors often arise from reusing variable names. If something does not appear to work correctly, try restarting this notebook and re-run all cells in order!

Conclusion

This activity introduced you to how raw image data is processed to produce accurate, full-color images. You explored key concepts in image reconstruction and learned how early image processing steps impact the quality and reliability of visual data in computer vision.